Overclocking the p-bit: Stochastic Binary Neural Network Inference with Autocorrelated sMTJ Randomness

Manan Gupta

Abstract*—Stochastic magnetic tunnel junctions (sMTJs) produce physically random bits at about 0.03 pJ per bit, five to six orders of magnitude less energy than software pseudorandom generation. They are therefore promising entropy sources for probabilistic-bit (p-bit) neural hardware. The usual design rule requires a sampling interval of at least twice the mean dwell time so that successive bits are uncorrelated. Existing systems follow this rule, but its application-level cost is unknown. We treat the ratio of sampling interval to correlation time as a continuous design parameter and simulate stochastic binary-network classification on MNIST and Fashion-MNIST using telegraph sMTJ neurons. The operating mode matters more than the sampling rate. In settle mode, a measured per-input relaxation of about four correlation times makes accuracy nearly independent of rate: with 32 samples, sampling $200\times$ faster than the rule costs less than 0.7 percentage points. In continuous streaming, however, stale states carry activations from one input to the next, and single-sample accuracy falls to chance. Device-to-device dwell-time dispersion does not solve this problem. At a fixed central fluctuation rate, it modestly reduces accuracy. The more consequential choice is the population statistic held fixed in fabrication: fixing mean rather than median correlation time multiplies every device rate by $e^{\sigma_\Delta^2/2}$, a $3\times$ shift at $\sigma_\Delta = 1.5\,k_BT$. Finally, correlation-aware training, which backpropagates through correlated activations, recovers much of the streaming loss (28% to 85% at the harshest setting) for a 3.3-point cost at nominal speed. It extends the accuracy–throughput Pareto frontier to $2\times$ the rule-compliant optimum at 97% accuracy, $4\times$ at 95%, and $8\times$ at 90%.

***Index Terms*—probabilistic computing, p-bit, stochastic magnetic tunnel junction, binarized neural networks, random telegraph noise, energy-efficient inference**

I. Introduction

GENERATING randomness is expensive on conventional hardware: pseudorandom generation at Gb/s rates costs about 30 nJ per bit on CPUs and 3 nJ per bit on GPUs [1]. A stochastic magnetic tunnel junction (sMTJ) is a nanomagnet whose free layer thermally switches between two states. It produces physically random bits at about 0.03 pJ per bit [1]. This roughly 10^6-fold energy gap motivates probabilistic-bit (p-bit) computing [2], [3]. In a p-bit system, the sMTJ is the stochastic neuron rather than a separate random-bit generator: an input current biases its time-averaged state to $\langle m\rangle = \tanh(I)$ [2], the activation used by stochastic binary neural networks (SBNNs).

An sMTJ also has its own switching timescale. Reading it faster than it switches returns repeated, stale states rather than independent samples. Daniels *et al.* found that adjacent samples become uncorrelated only when the sampling interval exceeds about twice the *mean dwell time* τ_d [4]. This "2τ rule" is commonly treated as a hard constraint. Demonstrated systems oversample the relaxation time by large factors [5], and random-bit quality is assessed with statistical test suites [1], [6]. Prior work has examined the timing trade for sampling machines: autonomous Ising machines require synapses to respond faster than neuron fluctuations [7], and in situ Boltzmann learning requires a learning time constant longer than the MTJ fluctuation time [8]. What remains unclear is the application-level cost. No published study continuously maps sampling ratio to feedforward classification accuracy, separates the operating modes in which correlation matters, or tests how much training can recover. This paper addresses those questions with three contributions:

- **Operating mode matters more than rate.** We model sMTJ neurons as input-biased telegraph processes with an exact discrete-time propagator and the stationary p-bit activation law. We sweep $r_0 = \Delta t/\tau_{\mathrm{corr}}(0)$ across four decades. In settle mode, we measure rather than assume the required relaxation interval: a settle of about $4\tau_{\mathrm{corr}}$ makes classification nearly rate-independent. At $r_0 = 0.02$, which is $200\times$ faster than the rule, 32-sample accuracy falls only from 97.7% to 97.1%. In continuous streaming, stale activation states cross input boundaries and single-sample accuracy reaches chance (10%) below $r_0 \approx 0.05$. Increasing samples by $32\times$ shifts this threshold by roughly one order of magnitude in r_0. The same collapse appears, and is worse, for a bias-independent-rate variant, indicating that the cause is generic staleness rather than the Arrhenius model.
- **The centering convention matters more than dispersion.** Whether device-to-device dwell-time dispersion appears helpful or harmful depends on the population statistic held fixed during fabrication. Because dwell time varies exponentially with barrier offset, fixing the *mean* correlation time instead of the *median* device multiplies every rate by $e^{\sigma_\Delta^2/2}$, or $3.1\times$ at $\sigma_\Delta = 1.5\,k_BT$. This uniform rate shift explains nearly all of the apparent recovery from 72% to 94% at $r_0 = 0.05$. At matched central rates, dispersion costs 1–5 accuracy points because the slow tail dominates. Device specifications should therefore state the central fluctuation rate explicitly; dispersion around it is a smaller effect.

M. Gupta is with The Harker School, San Jose, CA, USA (e-mail: mnn@yogins.com).

- **Correlation-aware training extends the Pareto frontier.** Training with activations sampled from the correlated telegraph process, with state carried across mini-batches, recovers most of the streaming loss: at $r_0 = 0.02$ and T=32, accuracy rises from 28% to 85%. This costs 3.3 points at nominal speed; a milder version costs 1.2 points. When throughput includes the measured $\approx 4\tau_{\mathrm{corr}}$ settle time per input, the frontier reaches 2× the best rule-compliant operating point at 97% accuracy, 4× at 95%, and 8× at 90%. The two larger gains come from correlation-aware training.

II. Background and Related Work

A. p-bits and sMTJ entropy sources

A p-bit produces $m = \mathrm{sgn}\{\mathrm{rand}(-1, 1) + \tanh I\}$ on each query, so its mean is $\langle m\rangle = \tanh(I)$ [2]. Superparamagnetic MTJs with barriers near $1.4\, k_B T$ [1] implement this behavior natively. Their dwell time is $\tau = \tau_0 e^{\Delta/k_B T}$, where $\tau_0 \approx 10\,\mathrm{ps}$–$1\,\mathrm{ns}$ [2]. Reported fluctuation times range from 2–5 ns autocorrelation in easy-plane devices [9] and nanosecond-scale telegraph noise in in-plane junctions [10], to the slower devices used in early p-bit demonstrations [11]. One early system operated with millisecond-scale fluctuations. A recent heterogeneous CMOS-plus-nanomagnet system samples a roughly 20 ms relaxation time at 40× the rate [5].

B. Randomness quality, sampling, and neural inference

Robustness studies of MTJ neural hardware have mainly examined *static* imperfections, including sigmoid distortion from process variation [12], [13] and parameter spread in fabricated arrays with compensation [14]. Work on sMTJ generators typically evaluates random-bit streams with NIST tests [1], [6]. Other studies show that PRNG quality can have a threshold effect on MNIST training [15], and that measured sMTJ bitstreams can support stochastic-computing inference despite their native correlation [16]. Bit-level correlation has long been a primary variable in stochastic computing [17], [18]. Recent work also finds that averaging stochastic binary activations, and training through that average, improves p-bit network accuracy [19]. The theory is well established: autocorrelation reduces the effective sample size of a Monte Carlo average [20], and spiking networks can have an optimal noise level [21]. We study the missing combination: a continuous sampling-ratio sweep mapped to feedforward classification accuracy across operating modes, variability models, and training regimes.

III. Model

A. The sMTJ neuron as a biased telegraph process

Each hidden unit is a two-state fluctuator $s \in \{-1, +1\}$ with input-modulated Arrhenius escape rates

$$k_{-\to+} = f\, e^{+I}, \qquad k_{+\to-} = f\, e^{-I}, \tag{1}$$

where I is the dimensionless synaptic input and $f = f_0 e^{-\delta\Delta}$ the device attempt rate with per-device barrier offset $\delta\Delta$ (in $k_B T$). The stationary law is

$$P(s{=}{+1}) = \frac{k_{-\to+}}{k_{-\to+} + k_{+\to-}} = \frac{1 + \tanh I}{2} \equiv p, \tag{2}$$

because $\langle m\rangle = \tanh I$ and $\langle m\rangle = 2P(+1) - 1$ [2]. Thus each neuron has the p-bit activation law at every operating point; changing the sampling rate changes only its temporal structure. The correlation time is

$$\tau_{\mathrm{corr}}(I) = \frac{1}{k_{-\to+} + k_{+\to-}} = \frac{1}{2f \cosh I}, \tag{3}$$

so strongly biased neurons decorrelate faster. Correlation time and mean dwell time are different quantities. At I=0, $\tau_d = 1/f = 2\tau_{\mathrm{corr}}$, so the Daniels $2\tau_d$ rule [4] is $r_0 \geq 4$ in our units. At nonzero bias, the occupancy-weighted dwell-to-correlation ratio grows as $2\cosh 2I$; the corresponding dwell-time rule is therefore stricter for driven devices. We quote the rule at I=0:

$$r_0 \;=\; 2 f_0 \Delta t \;=\; \Delta t/\tau_{\mathrm{corr}}(0)\big|_{\delta\Delta=0}. \tag{4}$$

We assume each input remains constant during an inter-query interval, as in sample-and-hold operation between clocked layers. The exact two-state propagator then has no discretization error:

$$P(s_{t+\Delta t}{=}{+1} \mid s_t) = p + (\mathbb{1}[s_t{=}{+1}] - p)\, e^{-\Delta t/\tau_{\mathrm{corr}}}, \tag{5}$$

with $\Delta t/\tau_{\mathrm{corr}} = r_0 e^{-\delta\Delta} \cosh I$.

Equation (1) describes the energy-tilt behavior of barrier-type devices in the Arrhenius regime. Easy-plane sMTJs are designed to have a fluctuation rate that changes little with bias [1], [9]. We therefore also simulate a **bias-independent-rate variant**: its total rate is fixed at $2f$, so $\Delta t/\tau_{\mathrm{corr}} = r_0 e^{-\delta\Delta}$ while the stationary law remains the same. Device variability is modeled by a barrier offset $\delta\Delta \sim \mathcal{N}(\mu, \sigma_\Delta^2)$ that is fixed for each chip. Since dwell time depends exponentially on $\delta\Delta$, the choice of μ determines the population statistic held fixed as σ_Δ grows. Setting $\mu = 0$ fixes the *median* device and makes the mean correlation time grow as $e^{\sigma_\Delta^2/2}$. Setting $\mu = -\sigma_\Delta^2/2$ fixes the *mean* correlation time. We report both cases because either can describe a fabrication corner. Neither changes the stationary firing probability, which isolates the temporal effect of variability from static sigmoid distortion [12]–[14]. Our model omits read disturb, $1/f$ resistance noise, temperature drift, and behavior outside the two-state telegraph regime [9], [10].

B. Operating modes: settle vs. streaming

In **settle mode**, which models batch-style inference, an input is applied and the device relaxes for $t_s = \rho\, \tau_{\mathrm{corr}}(0)$ before the first of T reads. Consecutive reads are separated by Δt. A device left idle for a long time equilibrates to $p = 1/2$, not to the stationary distribution of the next input. In continuous operation, its initial state is instead the final state from the previous input. In both cases, settling moves the device toward $p_{\mathrm{eq}}(I)$ with time constant $\tau_{\mathrm{corr}}(I)$. Starting from idle, the first read has residual bias $-\frac{1}{2}\tanh(I) e^{-\rho \cosh I}$. We simulate this relaxation from the actual preceding state and sweep ρ to measure the needed interval. We charge the settle time to throughput: settle-mode throughput is $1/((T{-}1)r_0{+}\rho)$ images per τ_{corr} and cannot exceed $1/\rho$, regardless of read speed. In **streaming mode**, devices never idle, so their states persist across inputs; throughput is $1/(T r_0)$. For a single settled read, the inter-read interval is never used. Thus settle-mode

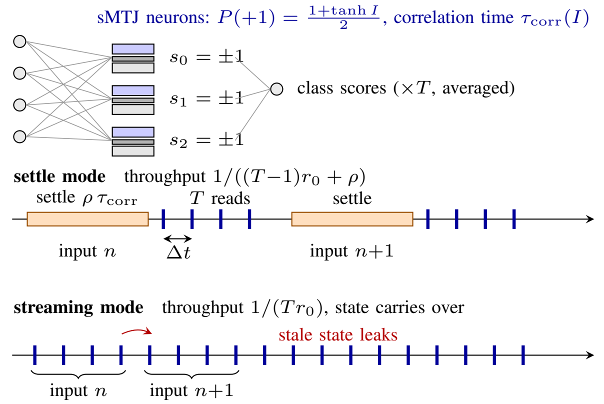

Fig. 1. SBNN with sMTJ neurons (top) and the two operating modes (bottom). Settle mode pays $\rho\,\tau_{\mathrm{corr}}$ of relaxation per input; in streaming mode devices are never idle and the previous input's activation pattern persists in the device states.

accuracy at $T{=}1$ depends only on ρ, and its flatness over r_0 is a protocol check. The measured settle requirement and the cost of correlated reads for $T{>}1$ are the substantive results.

C. Network and training

We use a $784{-}256{-}128{-}10$ MLP. Its hidden activations are stochastic and bipolar, and we resample them at every query, including at inference. This follows p-bit operation rather than the test-time deterministic approach of binarized networks [22], [23]. The input and readout layers are real-valued. Class scores average T stochastic passes. Baseline networks train with ideal i.i.d. randomness and a straight-through estimator [24]. Correlation-aware networks use the same training procedure, but sample hidden activations from the telegraph model while carrying device state across mini-batches.

IV. Experimental Setup

We evaluate MNIST [25] and replicate the main pattern on Fashion-MNIST [26]. The final 5,000 training images form the validation split for all model selection. We reserve the 10,000 test images for reported results. Each network trains for 30 epochs with Adam (10^{-3} and cosine decay). With ideal randomness, the MNIST reference networks reach 97.0% at $T{=}1$ and 97.8% at $T{=}32$. We sweep $r_0 \in [0.02, 10]$ and $T \in \{1, \ldots, 64\}$ with 3–5 thermal seeds per setting. The binomial standard error for 10,000 test images is ± 0.2–0.5 points; error bars show the standard deviation across seeds. We repeat ideal-randomness anchors over five independently trained networks and repeat key E2/E3 settings over four additional networks. Their agreement is within 0.4 points, with the largest difference near the collapse threshold. Correlation-aware results average three training seeds per configuration, except $r_0^{\mathrm{train}}{=}0.1$, which has one.

For streaming evaluation, we divide the test set into 100–200 independent sequential lanes. Each lane begins with unscored warm-up images from the training set, so evaluated images are in steady state. Accuracy is stable across lane counts:

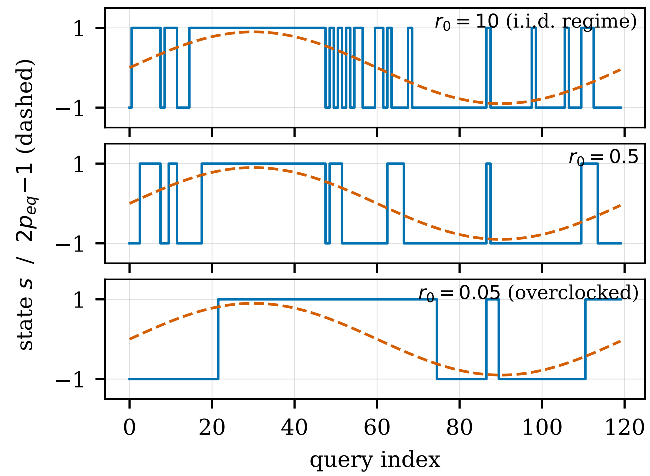

Fig. 2. Telegraph sMTJ neuron at three sampling ratios, driven by a slowly varying firing probability (dashed). At $r_0{=}10$ samples are effectively i.i.d.; at $r_0{=}0.05$ the device returns long runs of stale state.

72.6%, 72.2%, 72.2%, and 72.1% for 50, 100, 200, and 500 lanes at the same setting. Each result file records the simulator's git revision, and we reuse a run only when the source revision is unchanged. Unit tests validate the stationary distribution, autocorrelation decay at zero and nonzero bias, settle-relaxation distribution, and device variants against Eqs. (2)–(5). All experiments run on CPU in hours. The code, configurations, and per-run records are publicly available at https://github.com/mnn31/smtj-sbnn.

V. Results

A. Mode, not rate (E2)

Figure 3 gives the main result. In **settle mode**, single-read accuracy is 96.90% throughout the sweep when $\rho = 4$. This is expected: one settled read does not use the inter-read interval (Sec. III-B), so the flat curve validates the protocol rather than establishing a new result. The settle sweep measures the required relaxation. Single-read accuracy is 92.9%, 96.6%, 96.9%, and 96.8% for $\rho = 1, 2, 4, 8$, respectively. About $4\tau_{\mathrm{corr}}$ is sufficient, whereas $1\tau_{\mathrm{corr}}$ is not. The cost of correlation between repeated reads is small. At $T{=}32$ and $r_0 = 0.02$, or $200\times$ faster than the rule, accuracy decreases only from 97.73% to 97.12%. This residual loss follows the standard effective-sample-size penalty of correlated averaging [20]: stale within-input reads reduce T toward an effective T_{eff}, while the single-read accuracy remains high.

In **streaming mode**, steady-state accuracy collapses. Single-shot accuracy is 96.7% at $r_0 = 2$, 92.7% at 1, 68.6% at 0.5, 27.5% at 0.2, and 14.9% at 0.1. It reaches chance (about 10%) at $r_0 \leq 0.05$ because each input inherits the prior input's activation state. Increasing T by $32\times$ moves the 90% crossing by roughly one order of magnitude in r_0: the crossings are approximately 0.93, 0.33, and 0.09 for $T = 1, 8, 32$. Even with $T{=}32$, accuracy is only 28.1% at $r_0 = 0.02$. The bias-independent-rate variant has the same structure, but fails at sampling ratios about $2\times$ larger: at $r_0 = 0.1$, $T{=}32$ accuracy

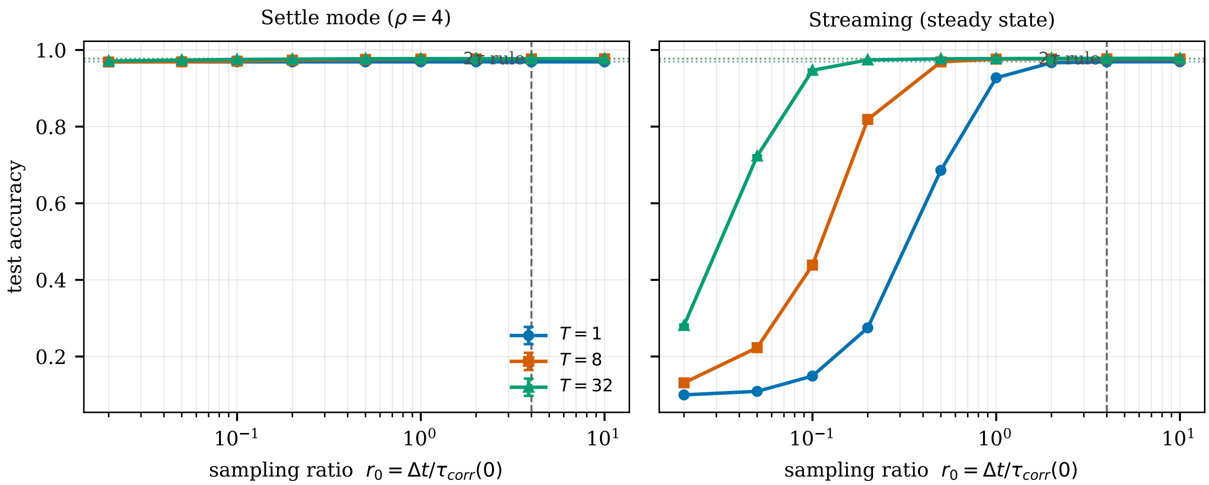

Fig. 3. Accuracy vs. sampling ratio r_0 (MNIST; mean ± std over thermal seeds; dotted: ideal-randomness anchors; dashed vertical: the $2\tau_d$ rule, $r_0 = 4$). **Left:** settle mode (ρ=4) is nearly rate-independent. **Right:** steady-state streaming collapses; T=1 reaches chance below $r_0 \approx 0.05$.

is 53.8%, compared with 94.7% for the Arrhenius model. The failure therefore comes from stale states, not from the particular rate parameterization. The $\cosh I$ rate increase in barrier-type devices helps because strongly driven neurons refresh faster. Fashion-MNIST shows the same pattern: at $r_0 = 0.02$, settle-mode accuracy at T=32 changes only from 88.2% to 87.0%, while streaming falls to 30.2% for T=32 and to chance for T=1.

B. *Centering, not dispersion, governs variability (E3)*

Because dwell time depends exponentially on barrier offset, adding dispersion has no unique meaning until the fixed population statistic is specified. Figure 4 shows two such cases at fixed r_0. If the *median* device is fixed, accuracy declines from 72.0% to 67.5% at $r_0 = 0.05$ as σ_Δ increases from 0 to 1.5. If the *mean* correlation time is fixed, accuracy instead rises from 72.0% to 94.4%. The latter improvement is primarily a centering effect. Relative to median centering, mean-τ_{corr} centering applies the same rate multiplier $e^{\sigma_\Delta^2/2}$ to every device, which is 3.1× at $\sigma_\Delta = 1.5$. Evaluating the dispersion-free baseline in Fig. 3 at the median-device effective ratio $r_0^{\text{eff}} = r_0 e^{\sigma_\Delta^2/2}$ predicts the mean-τ_{corr} result within a few points. For example, $r_0^{\text{eff}} \approx 0.15$ predicts at least 95%, versus the measured 94.4%. Dispersion itself contributes a 1–5 point penalty in both cases because the slow tail hurts more than the fast tail helps. For overclocked streaming, device specifications should identify the central fluctuation rate. "Mean τ_{corr}" and "median device" differ by $e^{\sigma_\Delta^2/2}$ in central rate, a larger effect than the dispersion the specification nominally constrains [14]. Although population-level diversity can be a computational resource [27], it is a secondary cost for this fixed network.

C. *Correlation-aware training (E4)*

Networks trained with the telegraph source learn representations that are more robust to stale states (Fig. 5). In streaming mode with T=32, training at $r_0^{\text{train}} = 0.2$ yields 84.6% accuracy at $r_0 = 0.02$, compared with 28.1% for the ideal-trained baseline. The 56.5-point recovery costs 3.3 points at nominal speed: 94.5% rather than 97.8%. The same cost appears when both models are evaluated at the rule-compliant $r_0 = 4$, so it does not result from the chosen comparison point. Training at $r_0^{\text{train}} = 0.5$ gives a smaller deep-overclock recovery but reduces nominal accuracy by only 1.2 points, to 96.6%. At T=1, training helps but does not prevent failure at the most aggressive rates. The $r_0^{\text{train}} = 0.2$ model reaches 84.1% at $r_0 = 0.2$, compared with 27.5% for the baseline, but no trained network avoids deep-overclock collapse at T=1. Training and averaging therefore complement each other. This extends multi-sample, or "sample-aware," training [19] to temporal correlation: the network learns robustness to dependence between samples, rather than simply benefiting from a larger sample count.

D. *Throughput and energy (E5)*

Settle-mode throughput cannot exceed $1/\rho = 0.25$ images per τ_{corr}, regardless of read speed. Streaming has no corresponding settle limit, but it suffers from stale-state failure. Correlation-aware training shifts that streaming curve (Fig. 6). We select correlation-aware points using only the validation split. Relative to the best rule-compliant point at each accuracy target, the frontier is 2.0× faster at 97% (streaming, T=2, $r_0 = 2$, 97.3%), 4.0× faster at 95% (correlation-aware single-shot streaming at $r_0 = 1$, 95.6%), and 8.0× faster at 90% (correlation-aware single-shot streaming at $r_0 = 0.5$, 92.5%). These values are threshold comparisons on a finite grid, not continuous optima. At the 90% target, the selected point gives up 4.4 points relative to the rule-compliant point that clears the target (96.9%). The comparison also assumes that readout electronics are not the bottleneck.

For a millisecond-scale device with $\tau_{\text{corr}} \sim 10\,\text{ms}$, the rule-compliant single-read optimum is about 25 classifications per second. The frontier supports about 100/s at at least 95% accuracy and about 200/s at at least 90% accuracy using the same devices. Randomness-generation energy depends on

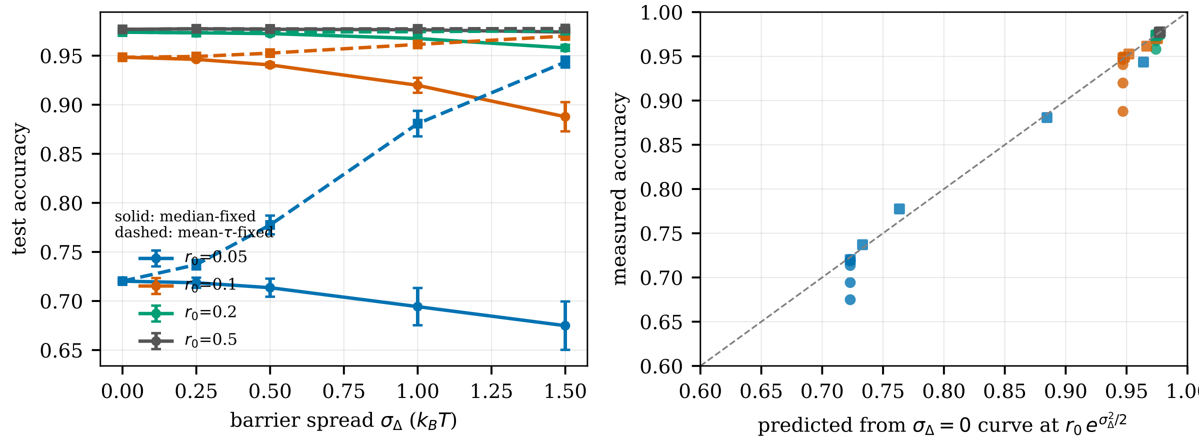

Fig. 4. **Left:** apparent effect of dwell-time dispersion depends on the fixed population statistic: median-fixed (solid) degrades; mean-$\tau_{\rm corr}$-fixed (dashed) appears to rescue (T=32, 5 chips, streaming). **Right:** the rescue is the centering's uniform rate offset in disguise: accuracy is predicted by the baseline ($\sigma_\Delta = 0$) curve evaluated at the median-device effective ratio $r_0\, e^{\sigma_\Delta^2/2}$ (mean-$\tau_{\rm corr}$) or r_0 (median), with dispersion contributing only a small residual penalty.

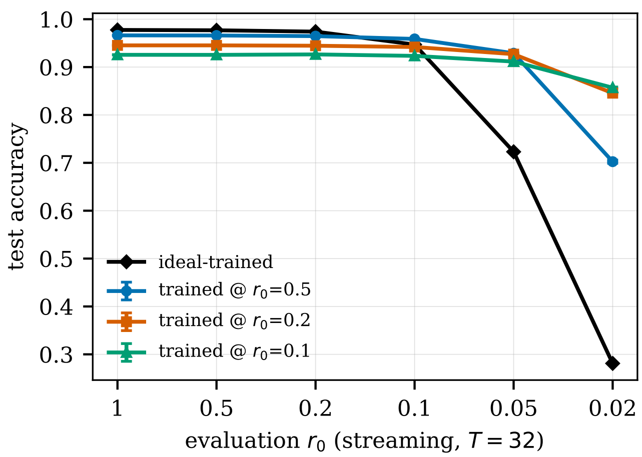

Fig. 5. Correlation-aware training flattens the streaming failure curve (steady-state, T=32; mean ± std over training and thermal seeds).

the number of bits, $384T$ per inference. At T=32, sMTJs use 0.37 nJ, versus 37–369 μJ for GPU or CPU PRNGs at the reported per-bit figures [1]. This five-to-six-order energy advantage does not depend on r_0; overclocking turns it into throughput by reducing the 2τ wait. Another way to decorrelate samples is to interleave k devices per neuron in round-robin order. That increases each device's effective sampling interval by k, moving along the same accuracy-versus-r_0 curves at $k\times$ the device area.

VI. Discussion

Design guidance. The $2\tau_d$ rule remains appropriate for TRNG bitstream certification [1], [6], but classification inference depends on workload. In settle-mode pipelines, devices can be read very quickly once each input receives about $4\tau_{\rm corr}$ of relaxation; that settle time becomes the throughput cost. Streaming has a real stale-state boundary. Correlation-aware

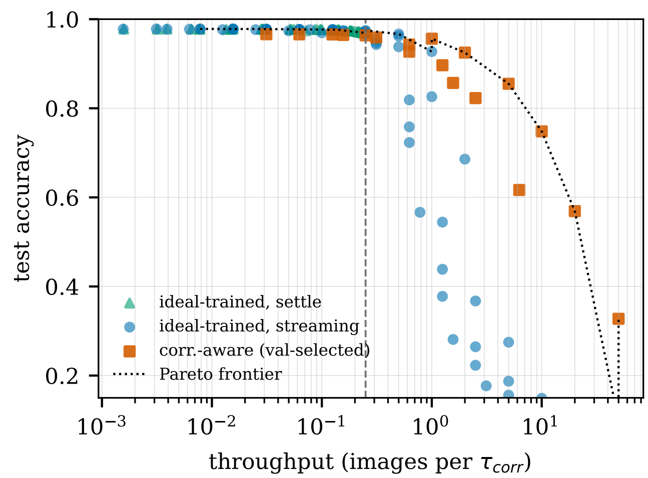

Fig. 6. Accuracy vs. physically-accounted throughput (settle mode pays its measured ρ=4 per input; dashed vertical: best rule-compliant single-read throughput r_0=4). Correlation-aware points use validation-selected training ratios.

training and sample averaging move that boundary, whereas fabrication uniformity does not. Specifications for overclocked devices should also identify the population statistic held fixed. Because dwell time depends exponentially on barrier height, the mean-$\tau_{\rm corr}$ and median-device conventions differ by $e^{\sigma_\Delta^2/2}$ in central rate. Finally, at matched $\tau_{\rm corr}$, barrier-type devices tolerate overclocking better than bias-independent-rate devices because strong inputs also refresh their neurons faster.

Limitations. This simulation uses an idealized two-state model. It does not include read disturb, retention drift, or $1/f$ noise [1]; it also uses small MLPs and MNIST-class benchmarks. The streaming experiments assume consecutive inputs are uncorrelated. Correlated input streams, such as video, could interact with staleness differently. The Pareto gains are threshold values on a finite grid, not continuous optima. Testing fabricated arrays such as the instrumented systems in [5], then extending the analysis to convolutional networks and temperature drift, would test these conclusions under more realistic conditions.

VII. Conclusion

Violating the sMTJ sampling rule has different consequences in the two operating modes. Batch-style inference can sample $200\times$ faster than the rule with less than one percentage point of loss, provided each input receives a measured settle of about $4\tau_{\mathrm{corr}}$. In steady-state streaming, stale device states drive accuracy to chance. Fabrication dispersion does not remedy that failure; its apparent benefit mainly reflects the chosen centering convention. Correlation-aware training recovers much of the loss. Treating sampling ratio as a design variable turns autocorrelation into throughput, reaching 2–$8\times$ the rule-compliant optimum depending on the accuracy target while retaining the roughly 10^6-fold random-bit energy advantage of p-bit hardware.

AI Disclosure

This work was developed with AI assistance under the author's direction: simulator and experiment implementation, and preparation of the manuscript including drafting, revision, and its related-work survey. An AI-assisted adversarial review was used to stress-test the methodology before submission. The author set the research questions and design, validated the results, and takes full responsibility for the content.

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